

Sequence and series

A **sequence** is an arrangement of any objects or a set of numbers in a particular order followed by some rule. whereas a **series** is the sum of all elements. **Sequence and series** are the basic topics in Arithmetic.

Difference Between Sequence and Series

Sequence	Series
In sequence, elements are placed in a particular order following a particular set of rules.	In series, the order of the elements is not necessary .
It is just a collection (set) of elements that follow a pattern.	It is a sum of elements that follow a pattern.
The Order of appearance of the numbers is important .	The order of appearance is not important .

Types of Sequence and Series

Some of the most common examples of sequences are:

1- Arithmetic Sequences

A sequence in which every term is created by **adding** or **subtracting** a definite number to the preceding number is **an arithmetic sequence**.

For example 1, 4, 7, 10, ... is an **arithmetic sequence**.

$1 + 4 + 7 + 10 \dots$ is an **arithmetic series**.

2- Geometric Sequences

A sequence in which every term is obtained by **multiplying** or **dividing** a definite number with the preceding number is known as **a geometric sequence**.

For example 1, 4, 16, 64, ... is **a geometric sequence**.

$1 + 4 + 16 + 64 \dots$ is **a geometric series**.

3- Harmonic Sequences

A harmonic sequence is a sequence where the sequence is formed by **taking the reciprocal** of each term of an arithmetic sequence.

For example 1, $\frac{1}{4}$, $\frac{1}{7}$, $\frac{1}{10}$, ... is **a harmonic sequence**.

$1 + \frac{1}{4} + \frac{1}{7} + \frac{1}{10} \dots$ is **a harmonic series**.

Sequence and Series Formulas

1- Arithmetic Sequence Formula

An arithmetic sequence is of the form:

$$a, a + d, a + 2d, a + 3d, \dots \text{up to } n \text{ terms}$$

where,

The first term is **a**, the common difference is **d**, the number of terms is **n**.

• Arithmetic Series Formulas

There are **different formulas** associated with an **arithmetic series** used to calculate the n^{th} term, sum, or the common difference of a given arithmetic sequence.

Formula 1: The arithmetic sequence formula to **find the n^{th} term** is given as,

$$a_n = a_1 + (n - 1)d$$

Example: Using the arithmetic sequence formula, find the 13th term in the sequence 1, 5, 9, 13...

Solution

$$\begin{aligned} a_n &= a_1 + (n - 1)d \\ &= 1 + (13 - 1)4 \\ \therefore a_n &= 1 + 48 = 49 \end{aligned}$$

Example: Find the first term in the arithmetic sequence where the 35th term is 687 and the common difference 14

Solution

$$\begin{aligned} a_n &= a_1 + (n - 1)d \\ 687 &= a_1 + (35 - 1)14 \\ \therefore a_1 &= 687 - 476 = 211 \end{aligned}$$

Formula 2: The **sum of first n terms** in an arithmetic sequence is calculated by using one of the following formulas:

$$S_n = (n/2) [2a_1 + (n - 1) d]$$

- (when we know the **first term** and the **common difference**)

$$S_n = (n/2) [a_1 + a_n]$$

- (when we know the **first term** and the **last terms**)

Example: Find the sum of the following arithmetic series: 3 + 7 + 11 + (up to 25 terms).

Solution

$$\begin{aligned} S_n &= (n/2) [2a_1 + (n - 1) d] \\ &= \left(\frac{25}{2}\right) [6 + (24)(4)] = 1275 \end{aligned}$$

Formula 3: The formula for **calculating the common difference** of an arithmetic sequence is given as,

$$d = a_n - a_{n-1}$$

2- Geometric Sequence Formula

Geometric sequence is of the form:

$$a, ar, ar^2, \dots, ar^{n-1}, \dots$$

where, the first term is **a**, the common ratio is **r**, the number of terms is **n**.

• Geometric Series Formulas

A **geometric series** is the sum of **finite** or **infinite** terms of a geometric sequence.

$1/2 + 1/4 + \dots + 1/8$ is a **finite geometric series** with the **first term**, $a = 1/2$ and the **common ratio** $r = 1/2$

$-4 + 2 - 1 + 1/2 - \dots$ is an **infinite geometric series** with the **first term**, $a = -4$ and the **common ratio** $r = -1/2$

The formulas for a geometric series include the formulas to find the **nth term**, the **sum of n terms**, and the **sum of infinite terms**.

Formula 1: The **nth term** of a geometric sequence is,

$$a_n = a r^{n-1}$$

Example: Find the 10th term of the geometric series $1 + 4 + 16 + 64 + \dots$

Solution

$$a_n = a r^{n-1}$$

Substitute $n = 10$, $a = 1$, and $r = 4$ in the above formula:

$$10^{\text{th}} \text{ term} = 1 \times 4^{10-1} = 4^9 = 262,144$$

Formula 2: The **sum of a finite** geometric series is

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{for } r \neq 1$$

$$S_n = \frac{a(r^n-1)}{r-1} \quad \text{for } r \neq 1$$

Example: Find the sum of the following geometric series:

$$1 + (1/3) + (1/9) + \dots + (1/2187)$$

Solution

$$a = 1, \text{ and } r = 1/3, \quad n = ??$$

In the given series, $n^{\text{th}} \text{ term} = 1 / 2187$

$$a r^{n-1} = 1 / 3^7$$

$$1 \times (1/3)^{n-1} = 3^{-7}$$

$$3^{-n+1} = 3^{-7} \quad \rightarrow n = 8$$

So we need to find the sum of the first 8 terms of the given series.

$$S_n = \frac{a(1-r^n)}{1-r} \quad \text{for } r \neq 1$$

$$S_8 = \frac{1 \times (1 - (1/3)^8)}{1 - 1/3} = \frac{1 - \frac{1}{6561}}{2/3} = 3280/2187$$

Formula 3: The **sum of an infinite** geometric series is

$$S_n = \frac{a}{(1-r)} \quad \text{for } |r| < 1$$

not defined for $|r| > 1$

Example: Find the sum of the following geometric series:

$$1 + (1/3) + (1/9) + \dots$$

Solution

$$S_n = \frac{a}{(1-r)}$$

$$a = 1, \quad \text{and} \quad r = 1/3$$

$$S_n = \frac{1}{(1-1/3)} = \frac{1}{2/3} = 3/2$$

Exercise (10-1)

1- Find a_{21} of an arithmetic sequence if $a_{19} = -72$ and $d = 7$.

Answer: $a_{21} = -58$.

2- Find the n^{th} term of the arithmetic sequence $-5, -7/2, -2, \dots$

Answer: $a_n = 3n/2 - 13/2$

3- Which term of the arithmetic sequence $-3, -8, -13, -18, \dots$ is -248 ?

Answer: -248 is the 50^{th}

4- Find the sum of the arithmetic sequence $-3, -8, -13, -18, \dots, -248$.

Answer: $S_{50} = -6275$.

5- Find the 35^{th} term in the arithmetic sequence $3, 9, 15, 21, \dots$

Answer: $a_{35} = 207$

6- Find the 125^{th} term in the arithmetic sequence $4, -1, -6, -11, \dots$

Answer: $a_{125} = -616$

7- Find the 10^{th} term of the geometric sequence $1, 3, 9, 27, \dots$

Answer: $a_{10} = 19,683$.

8- Find sum of the first 15 terms of the geometric sequence $1, 1/2, 1/4, 1/8, \dots$

Answer: $S_{15} = 1.99994$.

9- Find sum of the infinite geometric sequence $1, 1/4, 1/16, 1/64, \dots$

Answer: $S_{\infty} = 4/3$.

10- Calculate the sum of the finite geometric series if $a = 5$, $r = 1.5$ and $n = 10$.

Answer: $S_{10} = 566.65$.

11- Find the sum of the terms $1/9 + 1/27 + 1/81 + \dots$ to ∞ ?

Answer: $S_n = 1/6$